Drainage

Drainage is the **system of pipes and channels that remove wastewater, sewage, and rainwater from a building and transport it to disposal or treatment systems**.

It operates primarily through gravity, using pipes that are angled downwards to carry water away to a sewer or treatment facility. Among the most common issues are:

- Slow or blocked drains: most likely causes include grease and fat from cooking, leaves and other outdoor debris, tree root intrusions and foreign objects flushed down the toilet.
- Improper grading: the slope of the land directs water towards a property, not away from it.
- Overflowing gutters: often caused by blockages due to fallen leaves

The main objective of a drainage system is to collect and remove waste matter systematically to maintain healthy conditions in a building. Drainage systems are designed to dispose of wastewater as quickly as possible and should prevent gases from sewers and septic tanks from entering residential areas.

Technical words for parts of a drainage system.

- **Drain**-A pipe laid below ground level that uses gravity to carry waste water away from the building.
- **Gully**-The fitting on the drain that takes waste pipes from baths and basins but not Water closets.
- **Invert level**-the depth to the bottom of a drain pipe measured from the datum.
- **Manhole/inspection chamber**-the access point to the drainage system for inspection and cleaning.
- **Public sewer**-the large pipe laid below the roads or other public land which carries waste from building plots to the treatment works.
- **Soil drainage**-the pipework that takes the discharge from the soil and waste fittings such as the WC, bath and basin.
- **Soil vent pipe**-A pipe installed at the end of the drainage system for a building to expel gases from the sewage system.
- **Surface water drainage**-the water that flows from the roof and paved areas after rain. This water is considered clean and does not have to be treated at the sewage plant.
- **Trap**-a fitting that retains water in the drainage system. The trap prevents sewer gases entering a building or escaping at ground level.

1. TYPES OF WASTEWATER

In building services engineering, wastewater generated in buildings is generally divided into **three main categories**:

1. **Black Water or soil water**
2. **Grey Water or wastewater**
3. **Storm Water (Rainwater Runoff)**

Each type has **different characteristics, contamination levels, and disposal methods**.

i) Black Water

Black water is wastewater that **contains human waste and pathogens**, mainly from toilets and urinals.Black water comes from:

- Water closets (toilets)
- Urinals
- Bidets

Characteristics

- Contains **human waste (faeces and urine)**
- Has **high levels of bacteria and pathogens**
- Contains **organic matter**
- Has **strong odors**
- Requires **proper treatment before disposal**

Disposal Methods

In Kenya, black water is commonly disposed through:

- Municipal sewer systems
- Septic tanks
- Biodigesters
- Wastewater treatment plants

Must be carefully transported

ii) Grey Water

Definition

Grey water or wastewater is wastewater that **does not contain human waste**, but still contains soap, grease, food particles, and dirt. It comes from:

- Wash basins
- Showers
- Bathtubs
- Kitchen sinks

- Laundry machines
- Dishwashers
- Floor drains

Grey water typically:

- Contains **soap and detergents**
- Contains **grease and oils**
- Contains **food particles**
- Contains **less harmful bacteria compared to black water**

Although it is less contaminated, it is **still not safe for drinking**.

Reuse Potential

Grey water can sometimes be **recycled and reused** for: Landscape irrigation, Toilet flushing, Cleaning outdoor areas. This practice is becoming more common in **sustainable building design**.

iii) Storm Water (Rainwater)

Storm water is **rainwater that collects on roofs, paved surfaces, and open areas and must be drained away from buildings**.

Sources of Storm Water

Storm water comes from:

- Roof surfaces
- Balconies
- Driveways
- Parking areas
- Pavements

In building services engineering:

- **Untreated rainwater → Non-potable uses**
- **Treated rainwater → Can be used for potable purposes**

Drainage Components

Storm water systems include:

- Gutters
- Downpipes
- Surface drains
- Catch basins

- Stormwater drains
- Soak pits
- Stormwater sewers

Purpose

Storm water drainage prevents:

- Flooding
- Structural damage to buildings
- Soil erosion
- Water stagnation

Uses of Storm Water in Building Services

Storm water can be **collected from roofs and other surfaces** through gutters and downpipes and stored in tanks or directed to infiltration systems. Once collected (and sometimes treated), it can be used for several **non-potable purposes**.

1. Landscape Irrigation

One of the most common uses of storm water is watering gardens, flower beds, agricultural plots and lawns.

Since storm water is usually soft water (low mineral content), it is beneficial for plants.

2. Toilet Flushing

Storm water can be used for **flushing toilets in buildings**. Toilet flushing accounts for **a significant portion of indoor water consumption**. Using rainwater instead of treated potable water helps reduce water bills

Many **green buildings** integrate stormwater storage tanks connected to flushing systems.

3. Cleaning and Washing

Storm water can be used for **general cleaning activities**, including: Washing cars, cleaning pavements and walkways, washing outdoor areas and cleaning construction equipment

4. Construction Activities

Storm water can be used during **construction projects** for: Concrete curing, dust suppression, cleaning construction tools and soil compaction

5. Groundwater Recharge

Storm water can be directed to the ground through: Soak pits, Infiltration trenches. This helps to replenish underground aquifers, maintain groundwater levels and reduce surface flooding

6. Cooling Systems in Buildings

In some large buildings or industries, stored rainwater can be used in Cooling towers, HVAC systems and Equipment cooling

7. Fire Fighting Storage

Storm water can be stored in **dedicated tanks** and used as a **backup water source for fire protection systems**, including: Fire hose reels and sprinkler systems

This improves **building safety and resilience**.

8. Environmental Control

- Retention ponds
- Permeable pavements
- Green roofs

Main Components of Drainage

1. **Waste Pipes**
    - Carry water from sinks and basins
2. **Soil Pipes**
    - Carry waste from toilets
3. **Vent Pipes**
    - Release sewer gases
    - Maintain air pressure in pipes
4. **Floor Drains**
5. **Inspection Chambers / Manholes**
6. **Sewer or Septic Tank Connection**

Flow Direction

Water flows:

Fixtures → Waste Pipes → Soil Pipes → Manholes → Sewer / Septic Tank

Usually by **gravity**.

2. PRINCIPLES OF SOIL DRAINAGE.

Since drains laid below ground are fairly inaccessible, their operations must be as trouble free as possible. To achieve this the following principles, need to be followed in their construction**:**

i. Ensure that the drains are watertight so that waste does not leak out and contaminate the ground.
ii. Clear away obstructions within the pipes such as surplus mortar at the joints.
iii. Lay drains on even gradients to make sure that the water carries solid matter away smoothly.
iv. Lay drains in straight lines. If a drain changes direction, then insert an inspection chamber.

v. Construct a manhole if several pipes come together to join a single drain run.
vi. Construct inspection points that are less than 45 meters apart. Construct manholes at 90-meter intervals on straight lines.
vii. Construct a manhole at the boundary of the property before the drain joins the public sewer.
viii. Use soil drains that are at least 100mm in diameter.
ix. Surround drains under buildings in at least 150mm of concrete.
x. Backfill trenches for drains near or below foundation level with concrete.
xi. Join branch drains to the main drain at an oblique angle.
xii. Insert a trap at every inlet to a drain other than a soil pip

3. DRAINAGE SYSTEM

There are three main types of drainage systems:

i. The Combined System (Most Common in Established Urban Areas)

Historically, major towns like Nairobi, Mombasa, and Kisumu were developed using combined systems where a single pipe network carries both foul water (sewage) and surface water (stormwater).It was simpler and cheaper to install during early urban planning phases as it minimizes pipe lengths, but it places an unnecessary burden on the sewage treatment works, in that rainwater does not require treatment

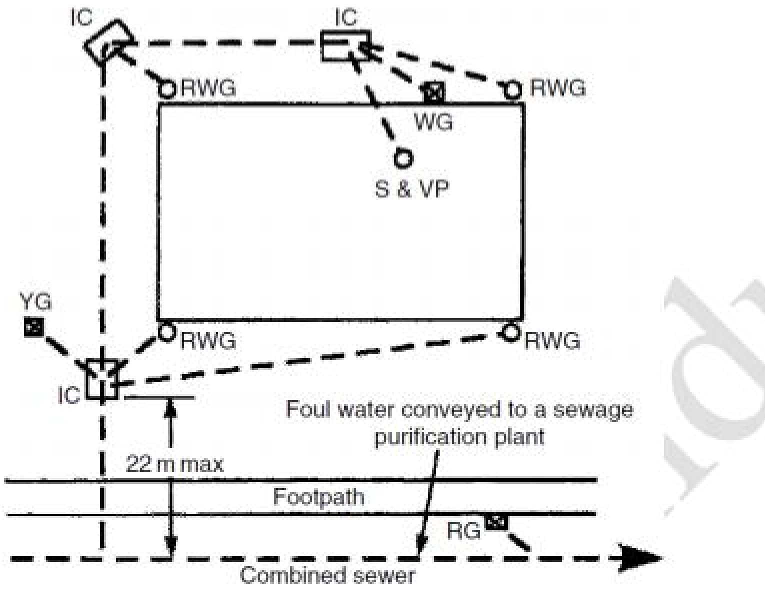
These systems are currently heavily overloaded due to rapid urbanization. During heavy rainy seasons in Kenya, this often leads to "Combined Sewer Overflows," where a mixture of rainwater and untreated sewage floods streets or overflows into rivers like the Nairobi River.

Advantages:

- Cost-Effective: Requires a single infrastructure, reducing initial investment.
- Efficient in Light Rain: Suitable for regions with frequent light rainfall.

Challenges:

- Pollution Risks: CSO can lead to water pollution as untreated sewage is discharged into natural water bodies during heavy rains.
- **Overflow Risk:** During the heavy rainy seasons experienced in Kenya (such as the "Long Rains"), these systems are prone to surcharging, where the volume of rainwater exceeds the pipe capacity, potentially causing backups.
- Limited Treatment: During heavy rain, treatment facilities might be overwhelmed, reducing the efficiency of wastewater treatment



The combined system

RWG-Rainwater gully
IC- Inspection chamber
WG- Waste gully
YG- Yard gully
RG- Road gully

S&VP- Soil and Vent pipe

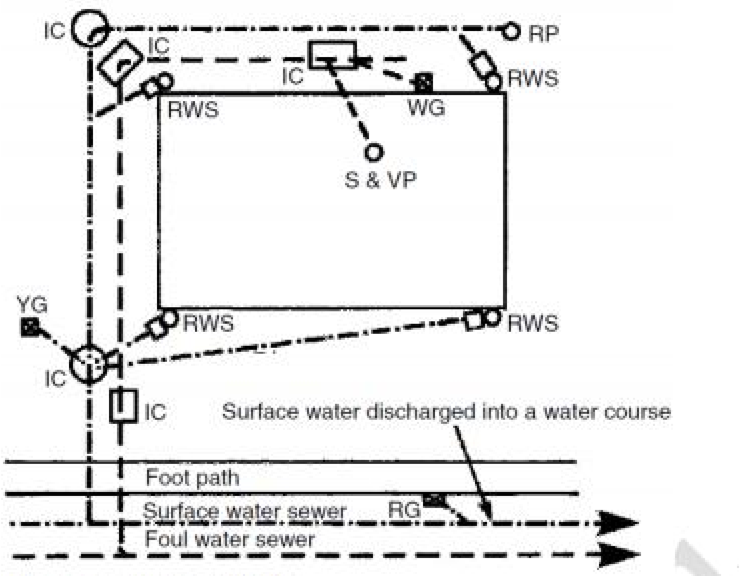
RP- Rodding Point

ii. The Separate System (The Modern Stan

The National Building Code 2024 and the National Construction Authority (NCA) now strictly mandate that "stormwater shall not enter a sewer installation."

This system uses a separate foul water drains that lead to a sanitary sewer. The rainwater from roofs and other surfaces is conveyed in a separate surface water drain into a surface water sewer. This system is relatively expensive to install.

- Where it's found: Newer residential estates (such as those in Syokimau, Ruiru, or parts of Karen) and modern commercial developments.
- How it works: Rainwater is directed to open concrete drains or soakaways, while sewage is directed to a separate sewer line or a septic/biodigester system. This is considered the most sustainable system for Kenya’s climate.



The separate system

Advantages:

• Reduced Pollution: Separation minimizes the risk of contaminating natural water bodies with untreated sewage.

• Efficient Treatment: Allows focused treatment of wastewater, improving overall water quality.

Challenges:

• Infrastructure Costs: Maintaining two separate systems can be more expensive.

• Limited Capacity: Stormwater pipes may be underutilized during dry periods.

iii. **Surface Drainage (Common in Rural & Peri-Urban Areas)**

This is the natural or artificial removal of excess water from the surface of the land. It involves the collection and conveyance of water(primarily from precipitation (rain) or snowmelt) to a suitable disposal point such as a river, stream, soakaway, or municipal storm sewer.

Effective surface drainage is critical to prevent soil erosion, ponding (standing water), and structural damage to foundations

Surface drainage relies on two primary physical principles:

- **Gradient (Slope):** Land must be graded or contoured so that gravity pulls water away from buildings and toward collection points. A common residential grade is a 2% slope (a 2-unit drop for every 100 units of distance).
- **Collection Points:** These are the physical structures designed to "catch" the moving water and transition it into a managed system.

An effective surface drainage system consists of several integrated parts:

Open Channels or Ditches: These are shallow, sloped trenches used to intercept and carry water. In urban planning, these often take the form of concrete **Invert Block Drains (IBDs)** found alongside roads. Grass-Lined Swales: Shallow, wide ditches covered with vegetation. These are aesthetically pleasing and help filter pollutants as water soaks into the soil.

- **Gutters and Downspouts:** These collect rainwater directly from the roof. The downspouts then discharge the water onto the ground (via splash blocks) or into an underground drainage network.
- **Catch Basins (Gully Pots):** These are buried masonry or plastic boxes with a grate on top. They collect surface water while allowing heavy sediment and debris to settle at the bottom, preventing clogs in the downstream pipes.
- **French Drains:** A perforated pipe buried in a trench filled withgravel. While it handles some sub-surface water, its primary role in surface drainage is to provide a "sink" for water to enter the ground quickly in areas prone to ponding.
- **Channel Drainage (Trench Drains):** Unlike point drainage, which collects water at a single spot, channel drainage collects water along a continuous line.

Surface vs. Sub-Surface Drainage

It is important to distinguish between these two:

- **Surface Drainage:** Deals with water *before* it enters the soil. It uses open channels, grading, and inlets.
- **Sub-Surface (Subsoil) Drainage:** Deals with water *after* it has soaked into the ground. It uses perforated pipes (like weeping tiles) located at or below the level of the building's footings to manage the water table

iv) Subsoil drainage

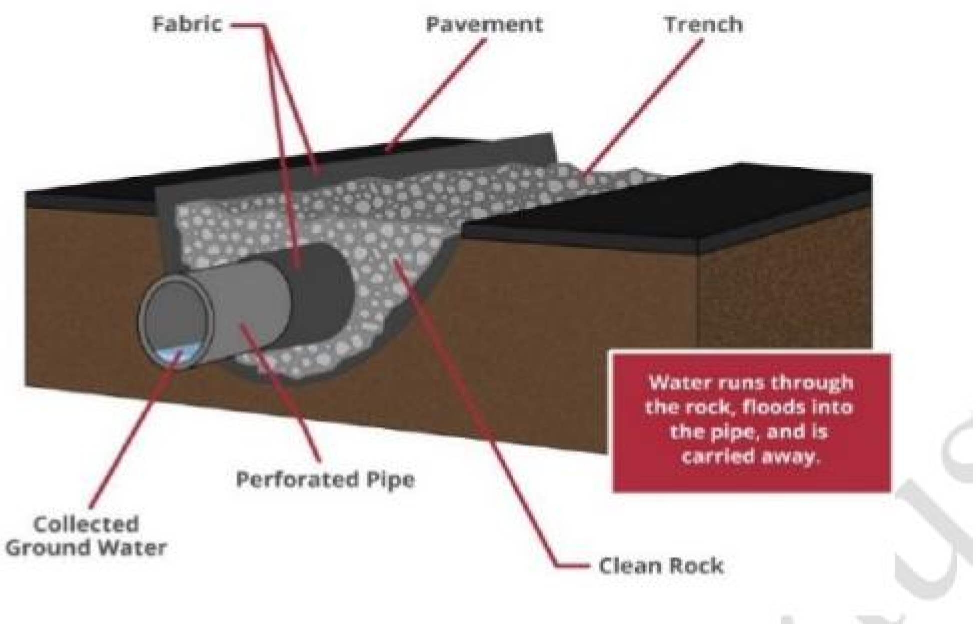
Ideally, buildings should be constructed with foundations above the subsoil water table. Where this is unavoidable or it is considered necessary to generally control the ground water, a subsoil drainage system
is installed to permanently lower the natural water table.
Various ground drainage systems are available, the type selected will depend on site conditions. Some of these systems include the following:

- French Pipes.
- Mole drains.
- Interceptor drains.
- Ground water pump

a) **French Drains / Soak Pits:** Common in private compounds where there is no municipal sewer connection. They allow water to seep naturally into the ground through gravel-filled trenches.

- **Structure:** It consists of a trench filled with rounded gravel and a perforated pipe (usually PVC or flexible HDPE) at the bottom.

- **The "Filter" Concept:** In professional installations, the entire gravel bed is wrapped in a **geotextile fabric** (filter fabric). This allows water to pass through while preventing fine soil particles from entering and clogging the pipe.
- **Application:** These are most commonly used around the perimeter of building foundations (often called "footing drains") to prevent water from seeping into basements or crawlspaces.

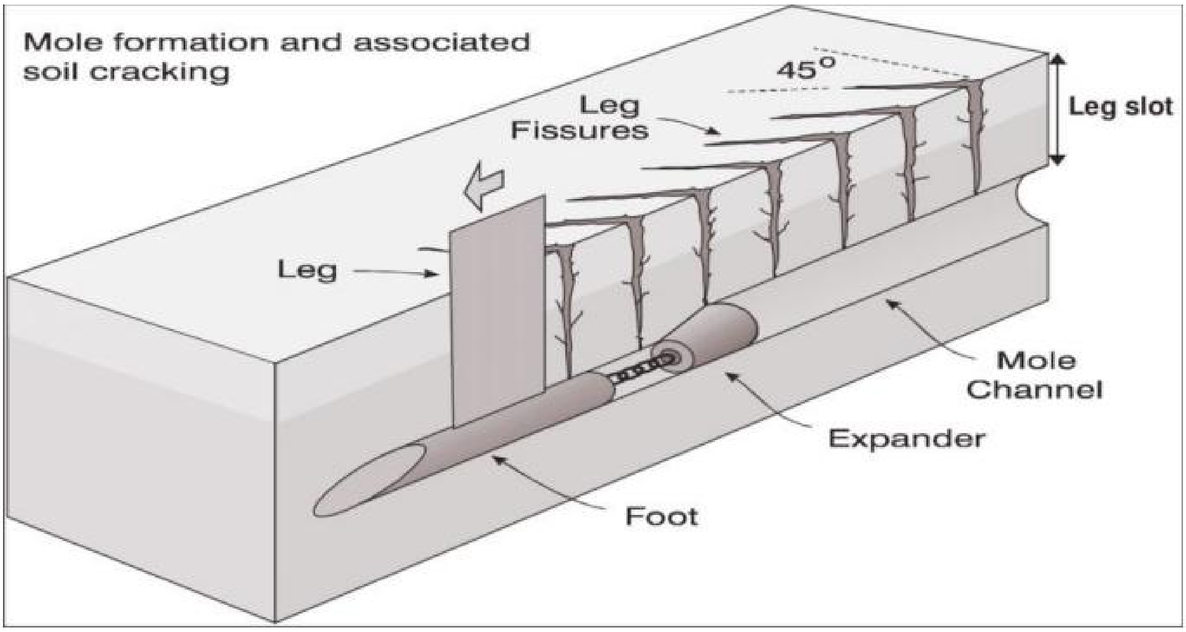


b) Mole Drains

Mole drains are a unique, "pipeless" form of sub-surface drainage typically used in heavy clay soils (like the "Black Cotton")

They do not drain groundwater but remove water as it enters from the ground surface

- **How they are made:** A tractor pulls a "mole plow"—a vertical blade with a cylindrical "bullet" at the bottom. This bullet creates a continuous, unlined horizontal tunnel (the "mole channel") about 400mm to 600mm below the surface.
- **Mechanism:** The clay soil is packed tight by the bullet, forming a temporary pipe that allows water to flow through the hollow space.
- **Lifespan:** Because they have no actual pipe, they eventually collapse or silt up, typically lasting 2 to 5 years before needing to be "re-malled."



c) Interceptor drains/Cut off drains

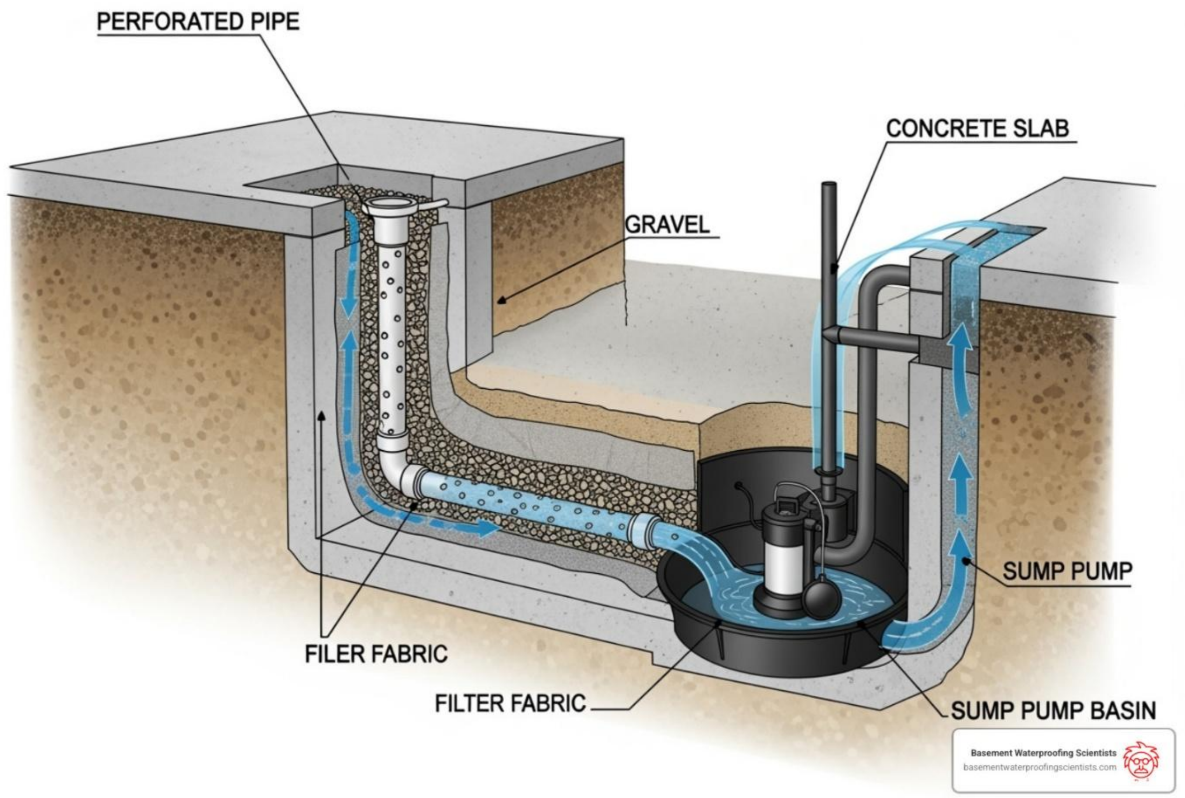
These are strategically placed to "intercept" groundwater that is moving laterally through the soil, usually on a slope. Meaning that the drains are placed at the base of slopes, typically where a steeper slope meets flats, to catch subsurface water flowing downward.

- **The Problem:** On sloped sites, water often moves sideways through the upper layers of soil. If a house is built on that slope, the water hits the uphill foundation wall.
- **The Solution:** An interceptor drain is installed **uphill** from the structure. It "cuts off" the water's path and carries it around the building to a lower discharge point.
- **Design:** They are usually deeper than standard surface drains and are filled with highly permeable stone to ensure the moving water enters the drain rather than continuing toward the building.

Interceptor drains can also be constructed beneath springs and spring lines to capture spring water.

c) Ground Water Pumps

These drain water from aquifers in order to maintain or lower the water table to an appropriate level below the surface of the earth. Pumping results in a drawdown of ground water away from the pump point; this effect is much less pronounced at deeper depths. Aquifer depth, soil type, water table height, and other factors will all affect how much of an impact there is



4. **ACCESS TO THE DRAIN**

Drain access may be obtained through a number of ways. Because drainage systems are largely hidden underground or behind walls, they are "out of sight, out of mind" until a problem occurs. We need physical access to drains for three primary reasons:

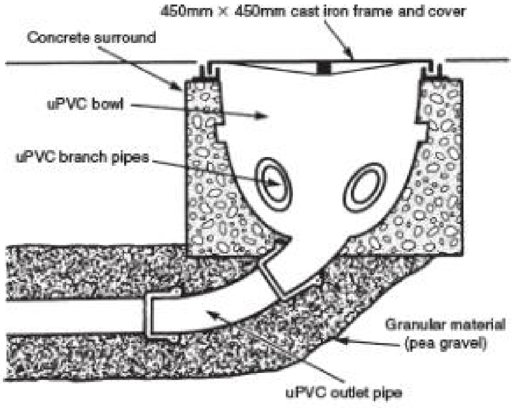
a) Inspection and Monitoring
b) Maintenance and Clearing Blockages (Rodding)
c) There is a change of Direction or Gradient

Pipe runs should be straight and access provided only where needed, i.e.:

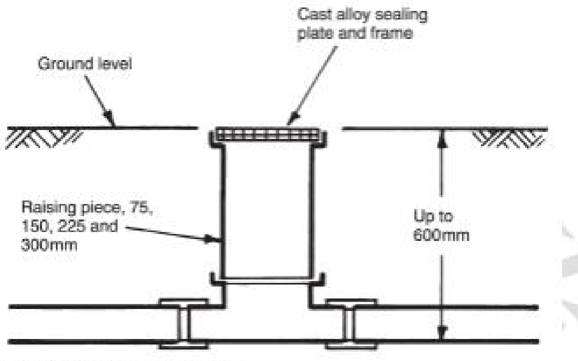
- at significant changes in direction
- at significant changes in gradient
- near to or at the head of a drain
- where the drain changes in size
- at junctions
- on long, straight runs.

a) Shallow Access chambers

Shallow access chambers or access fittings are small compartments similar in size and concept to rodding points, but providing drain access in both directions and possibly into a branch. They are an inexpensive application for accessing shallow depths up to 600 mm to invert. Within this classification, manufacturers have created a variety of fittings to suit their drain products. The uPVC bowl variation shown combines the facility of an inspection chamber and a rodding point



The Marscar access bowl

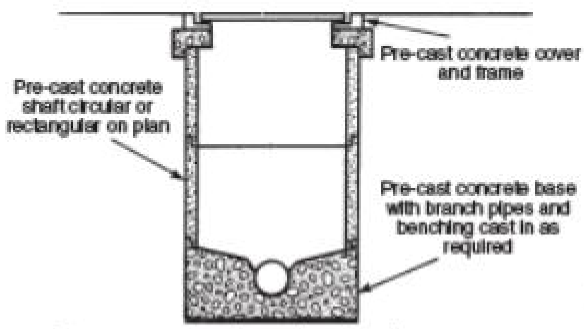


Clayware drain access fitting

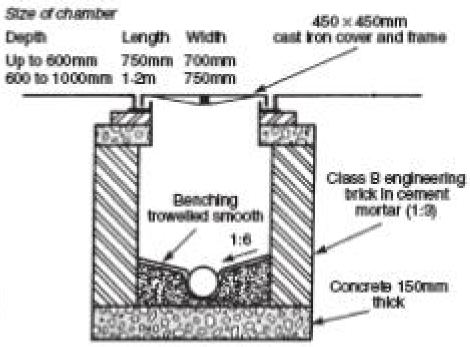
b) Inspection chambers

Inspection chambers are larger than access chambers, having an open channel and space for several branches. They may be circular or rectangular on plan and preformed from uPVC, precast in concrete sections or traditionally constructed with dense bricks from a concrete base. The purpose of an inspection chamber is to provide surface access only, therefore the depth to invert level does not exceed 1 m.

These are shallow chambers (usually up to 1 meter deep) with a removable cover. They are large enough for a person to look inside or insert tools, but not large enough for a person to enter.



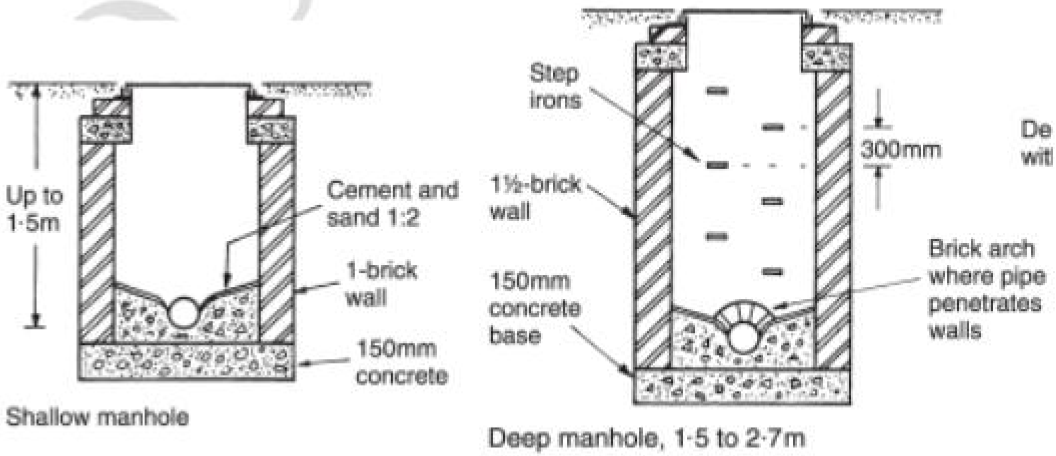
Pre-cast concrete inspection chamber



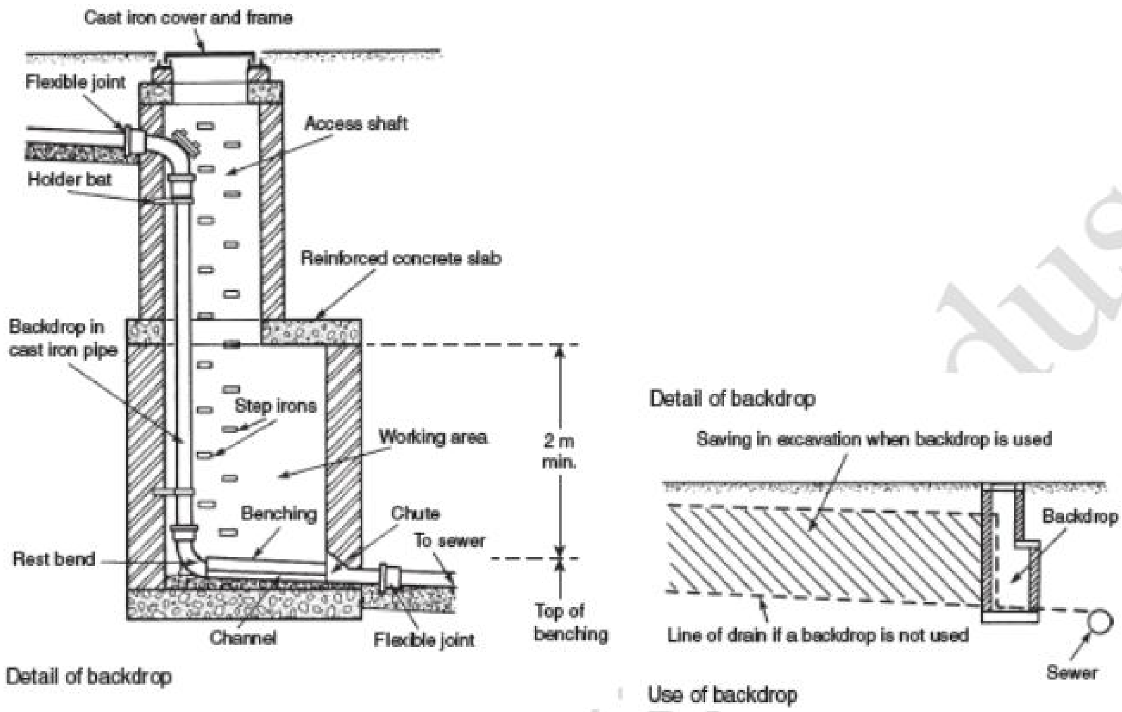
Brick inspection chamber

c) Manholes

The term manhole is used generally to describe drain and sewer access. By comparison, manholes are large chambers with sufficient space for a person to gain access at drain level. Where the depth to invert exceeds 1 m, step irons should be provided at 300 mm vertical and horizontal spacing. A built-in ladder may be used for very deep chambers.



Where there is a significant difference in level between a drain and a private or public sewer, a backdrop manhole may be used to reduce excavation costs. Backdrops have also been used on sloping sites to limit the drain gradient, as at one time it was thought necessary to regulate the velocity of flow. This is now considered unnecessary and the drain may be laid to the same slope as the ground surface. For use with cast iron and uPVC pipes up to 150 mm bore, the backdrop may be secured inside the manhole. For other situations, the backdrop is located outside the manhole and surrounded with concrete.



5. INSPECTION AND COMMISSIONING OF DRAINAGE INSTALLATIONS

Inspection and commissioning tests on drainage installations are carried out as follows.

I. Preliminary Inspection

During this phase, scrutinize the drainage system's design by reviewing plans and drawings. Assess layout, material adherence, intended functionality, and compliance with building codes, regulations, and environmental standards. Confirm permitting requirements.

II. Pre-Commissioning Checks

a) Visual Inspection

Assess the drainage system's physical condition, quality, and installation. Examine the entire system for defects, proper installation, secure connections, and sufficient pipe support. Scrutinize access points, including inspection chambers, manholes, and rodding eyes, for construction quality, secure covers, and effective sealing

b) Check Materials

This phase involves meticulous verification of materials used in the drainage system installation. Confirm that installed materials match design specifications. Key activities include checking for defects, ensuring materials are in good condition, and verifying the presence of required certifications.

c) Inspect Access Points

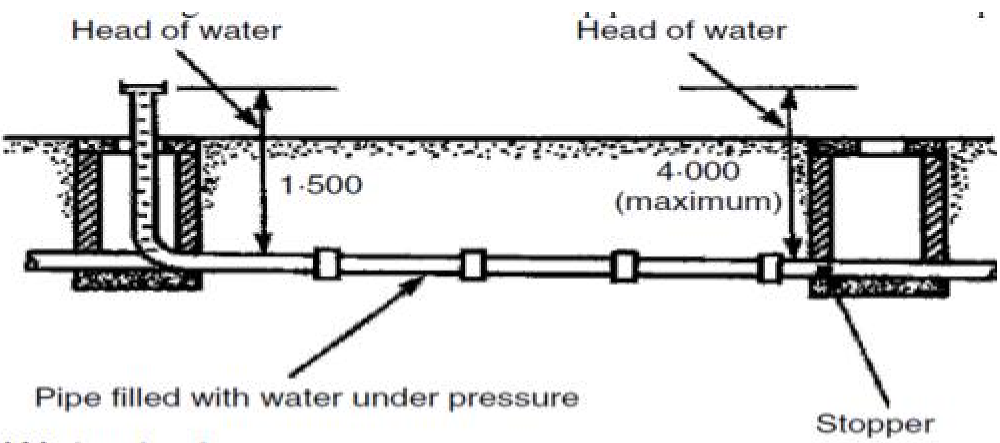
Crucial for confirming quality and functionality, this phase involves a detailed examination of access points. Inspection chambers, manholes, and rodding eyes are scrutinized for secure covers, proper construction, and effective sealing. Correct positioning is verified to ensure accessibility for maintenance and inspection
activities.

III. Leak Testing

a) Water Test

Leak testing involves pressurizing the drainage system with water to check for any leaks. It helps in ensuring the watertight integrity of the drainage system. The drainage system is filled with water, and its integrity is monitored. Any signs of water loss or leakage are identified, and corrective actions are taken promptly to address the issues.

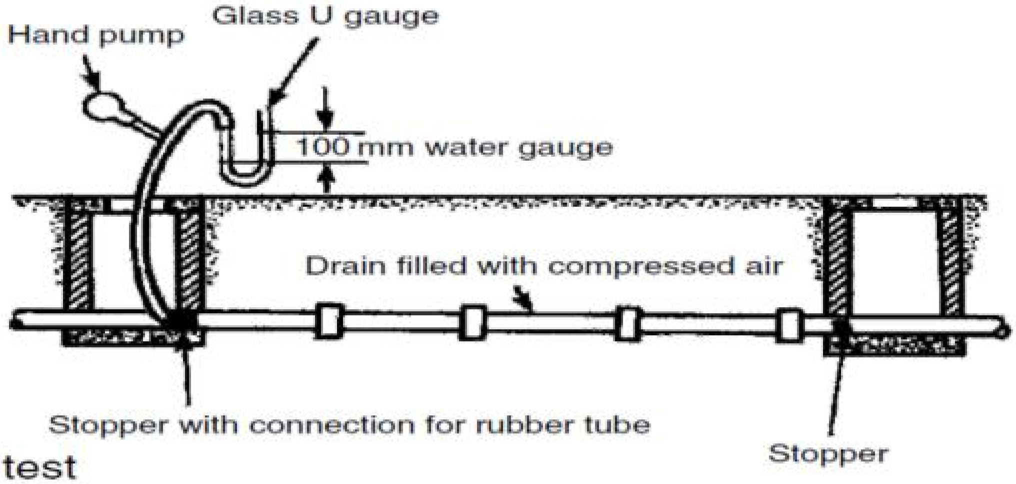
This test requires a purpose-made test bend with an extension pipe to produce a 1.5 m head of water. This should stand for 2 hours and if necessary, topped up to allow for limited porosity. For the next 30 minutes, maximum leakage for 100 mm and 150 mm pipes is 0.05 and 0.08 litres per metre run respectively



Water test

b) Air pressure test

In pressure-based drainage systems, pressure testing is conducted to assess the system's ability to withstand designated pressures. This involves subjecting the system to specific pressure conditions and ensuring that it can handle these pressures without any leaks or compromises in structural integrity. In this test the complete system will be tested on completion by filling the water seals and inserting air bags, expandable bungs, into the ends of stack pipes. A rubber hose is inserted into the vent stack through a WC water trap. The air pressure in the stack is hand-pumped up to 38 mm water gauge, measured on a U-tube manometer. This pressure must remain constant for 3 min without further pumping. Soap solution wiped onto joints will reveal leak locations.



Air test

c) Smoke test

Existing stacks can be tested by the injection of smoke from an oil-burning generator or a smoke cartridge, provided that it will not cause damage to the drain materials. This is less severe than the air test, as smoke pressure remains low and damage from the test itself is less likely. Suitable warnings must be given to the occupants of the building.

IV. Functional Tests

a) Flow Test

Functional tests simulate typical usage conditions to evaluate the drainage system's flow. Controlled flow is implemented to observe proper flow rates and identify any restrictions. This ensures that the drainage system effectively conveys wastewater without backups or impediments.

b) Check Drainage Appliances

This phase involves the verification of the functionality of fixtures and appliances connected to the drainage system. Sinks, toilets, and other fixtures are tested to confirm proper drainage. Any issues with drainage appliances are identified and rectified during this phase.

V. Inspection of Connections

a) Inspect Joints

The inspection of joints focuses on verifying the integrity of connections within the drainage system. Joints are examined for signs of leakage or inadequate sealing. It ensures that all connections are secure, properly sealed, and free from any potential issues.

b) Check Ventilation

This phase involves confirming proper ventilation within the drainage system. Vent pipes and air admittance valves are inspected to ensure correct installation. Adequate air circulation is verified to prevent airlock issues that may impede the system's functionality.

VI. Documentation and Record-Keeping

Compile Records

In this documentation phase, comprehensive records of all inspections and tests are compiled. Detailed notes on the results of the preliminary inspection, pre-commissioning checks, and subsequent tests are recorded. Any issues identified and the corresponding corrective actions are documented, providing a thorough account of the inspection process.

Any deviations from the original design are incorporated into the drawings to ensure an accurate representation of the installed system.

VII. Final Approval and Commissioning

a) Final Inspection

The final inspection phase ensures that all identified issues have been addressed. Verification of compliance with building codes and regulations is conducted. This phase confirms that the drainage system is ready for commissioning.

b) Commissioning

Commissioning officially declares the drainage system ready for operation. All necessary documentation is provided to relevant authorities, and the system is handed over to end users for regular use.

VIII. Post-Commissioning Follow-Up

a) Monitoring

A post-commissioning monitoring plan is implemented to detect any latent issues that may arise after the drainage system is put into regular use. Regular inspections are conducted to identify and address any emerging issues promptly.

b) Documentation

Post-commissioning activities are documented, including any maintenance, repairs, or additional inspections conducted after the system has been in operation. This documentation ensures a comprehensive record of the drainage system's ongoing performance

WASTE WATER MANAGEMENT SYSTEMS.

a) SOAK-AWAY

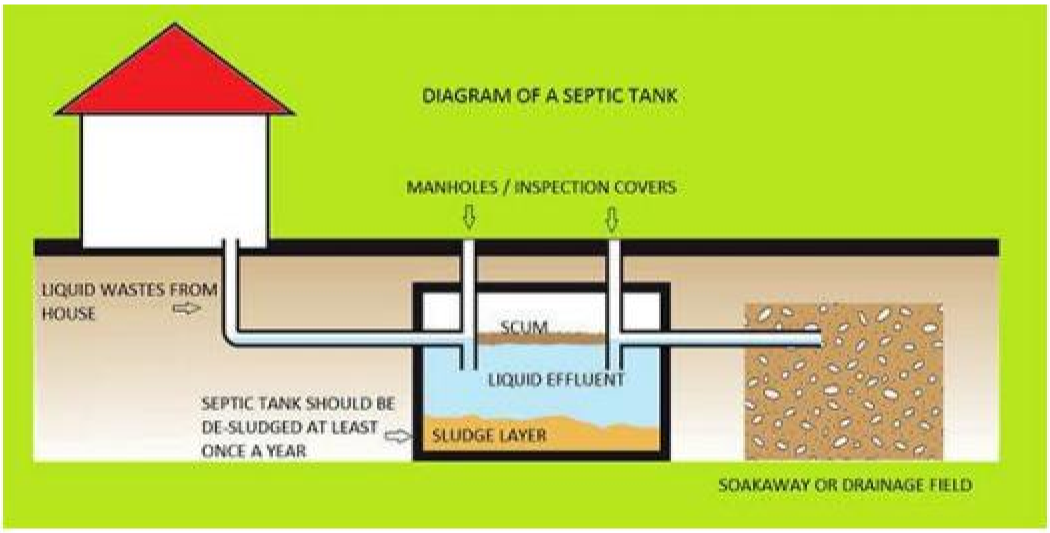
A soak pit (also referred to as a soak-away) is a covered, porous-walled chamber that allows water to slowly soak into the ground.

Unlike a sewer that carries water away to a treatment plant, a soakaway is designed to manage water locally by returning it to the ground. Soak pits serve as secondary chambers receiving effluent from a primary sanitation technology (e.g., septic tanks) for the treatment of wastewater (black water) and gradually allow the effluent to percolate into the surrounding soil.

Soak pits may also be used for treatment of greywater (water from personal hygiene, like showering and from kitchen areas). The soak pit can be lined with porous materials (rocks, gravel, sand, wood, or highly adsorbing minerals such as aluminum silicate). These materials may also provide foundational support to prevent collapse of the underground chamber.

Soak pits are sometimes left unlined and filled with coarse rocks and gravel. The rocks and gravel help prevent the chamber from collapsing and provide adequate space for the input of greywater and/or wastewater.

- **Collection:** Water from roof gutters, driveways, or sub-surface drains (like French pipes) is directed into the soakaway.
- **Infiltration:** Once inside, the water slowly seeps (percolates) through the walls and floor of the pit into the surrounding soil.



A soak away can either be:

- **Traditional (Filled):** A large pit filled with large, clean stones or broken bricks (hardcore). The gaps between the stones create the storage space.
- **Modern (Crates/Attenuators):** These use plastic "geocellular" crates. These are much more efficient because they are 95% void space, meaning a smaller pit can hold a much larger volume of water than a stone-filled one.

Whether a soak pit will function or not depends primarily on the permeability of the soil. Soil pores may become clogged with time and this can reduce the infiltration capacity of a particular soak pit. Seasonal variations in the water table can also affect the performance greatly, and a soak pit which works perfectly in the dry season may overflow at other times of year.

Soakaways location.

The soakaway should be:

• Downhill and the minimum safe distance from any drinking water source.

• At least 3 metres from the septic tank.

Soak pits are commonly between 2 and 5m deep and 1 to 2.5m in diameter. Wastewater entering the pit may soak into the surrounding soil through the sides and base of the pit. If the water has a high solids content, however, the base of the pit will quickly become blocked with silt and sludge. Where this occurs infiltration will only take place through the pit walls, therefore the base area is ignored when designing soak pits.

Advantages:

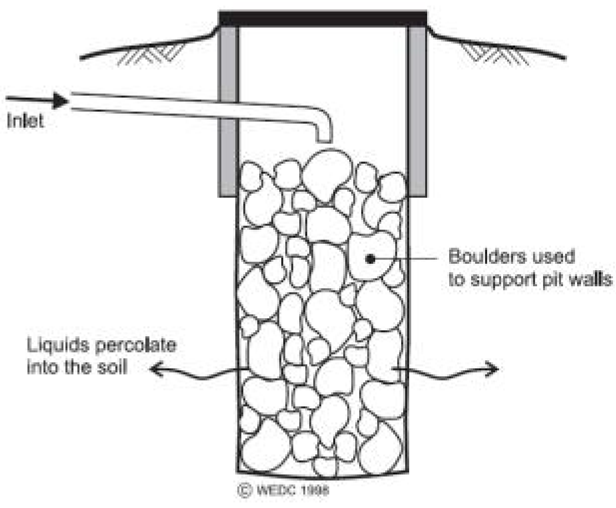
- **Environmental Sustainability:** It recharges the local groundwater table rather than sending runoff into over-burdened city pipes.
- **Cost-Effective:** It avoids the need for long runs of expensive piping to reach a public drain.
- **Quick and easy to construct**
- **Legal Requirement:** In many parts of Kenya, building regulations (and the National Building Code) mandate that stormwater must be managed on-site to prevent flash flooding in urban areas.

Critical Requirements for Success

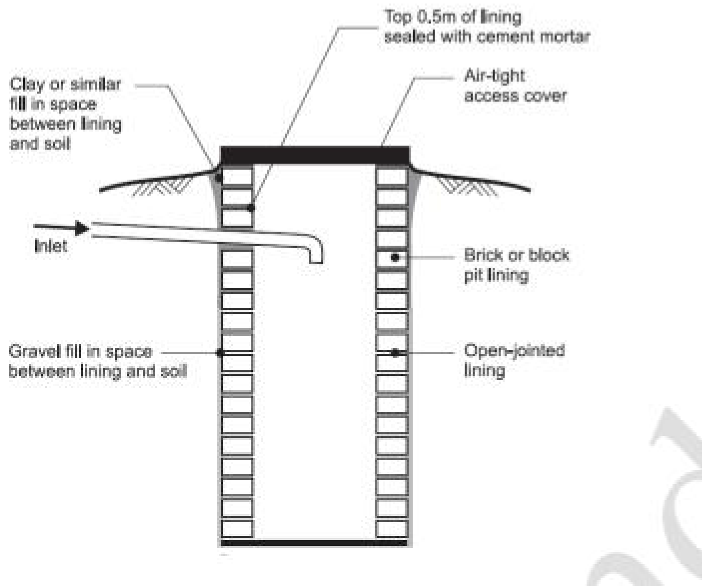
A soakaway is not suitable for every site. Two factors are vital:

1. **Soil Permeability:** If the soil is heavy clay (like "Black Cotton" soil), the water won't soak away fast enough, causing the pit to overflow. A **Percolation Test** is usually performed to see how fast the soil absorbs water.
2. **Location:** To protect the building, a soakaway must be placed at a safe distance, usually **at least 5 meters** away from any foundation, to prevent the soaking water from undermining the structure.

Unlined soakpit



Alternatively, the pit can be lined (as shown in Figure below). Any lining must be porous so that the wastewater can reach the soil surface. The top 0.5m of any pit must have a sealed lining in order to prevent the infiltration of rainwater.



The size of a soak pit depends on the volume of liquid to be disposed of and the type of soil in which the pit is excavated. It may be calculated by using the following process:

Table 1. WHO Suggested Infiltration Capacities (Ref. C.)

Type of soil	**Infiltration Capacity. (L / m² per day) (SIR)**
Coarse / medium sand	50
Fine sand, loamy sand	33
Sandy loam, loam	25
Porous silty clay / porous silty clay loam	20
Compact silty loam, compact silty clay loam and non-expansive clay	10
Expansive clay	<10

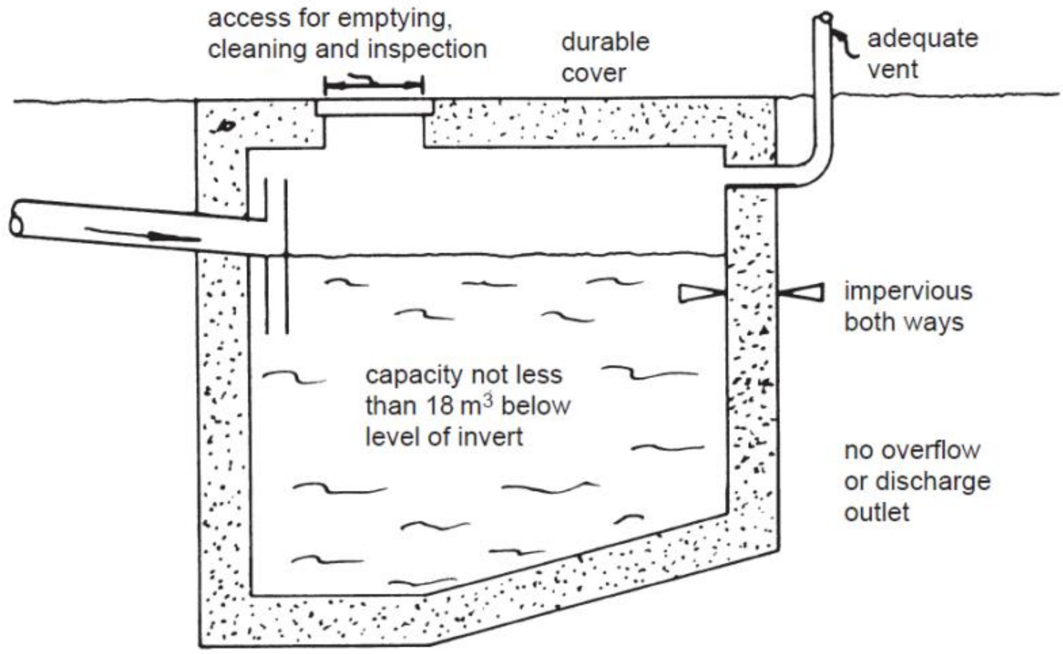
1. Calculate the surface area of pit wall required for infiltrating the wastewater:
   Pit wall area (m^2) = daily wastewater flow (litres) ÷ soil infiltration rate (Table 4.3)
2. Choose a pit diameter.
3. Calculate the depth of pit required to dispose of all the liquids:
   Depth of pit required = pit wall area ÷ (π x pit diameter)
4. Add 0.5m (lined depth) to calculate the total pit depth needed.

Worked example: A soak pit is required to dispose of 500 litres per day in a sandy loam soil (infiltration rate = 25 litres/m2 /day: see WHO Table above). There is space for a pit of 2m diameter only.

a) Surface area of pit = Daily wastewater/soil infiltration rate
   =500l/25l/m2
   =20 SM
b) Pit depth
   3.14*2*d=20
   D=20/(3.14*2)
   =3.18m
c) Total depth= pit depth+ lined depth
   =3.18+0.5
   =3.68m

b) CESSPOOL

cesspool (sometimes spelled cesspit) is a sealed underground tank used for the temporary collection and storage of raw sewage and liquid waste. Unlike a septic tank or a biodigester, a cesspool performs **no treatment** of the waste. It is simply a holding tank designed for collection and storage of sewage or other waste for short periods of time. Because the tank is completely sealed, no liquid is supposed to escape into the surrounding soil (unlike a soakaway or a septic drainage field). Once full, the contents must be vacuumed out by an exhaust tanker (commonly known in Kenya as a "honey sucker") and transported to a municipal treatment plant for disposal. This makes the overall maintainace expensive due to frequent emptying.



Cesspits should be manufactured to be watertight (e.g., lined with concrete or other non-permeable material) to fully contain the pathogens and reduce the possibilities of bringing odor back through the inlet pipes. Two or more vent pipes may be connected to the cesspit, and the pit may be either closed or slightly open (to vent away odors). Solids will settle and decay in the base as occurs within a septic tank. A watertight cesspit is expected to provide lower treatment and removal efficiencies than septic systems.

A cesspit is expected to serve only one household though it may be shared by families living in building that contains more than one toilet. The number of expected users determines the capacity of the tank to be installed which may range from 5 to 50 m3, depending on land availability. The number of users also determines the frequency of emptying the cesspit.

A cesspool should be sited:

- On a sloping ground away from and lower than nearby buildings
- Below and at least 7m from habitable parts of the building
- Within 30m of vehicle access
- At such levels such that emptying and cleaning can be carried out without creating a hazard for the building's occupants and without the contents being taken through a dwelling or place of work

c) SEPTIC TANKS.

Septic tanks are watertight chambers sited below ground level which receive excreta and flush water from flush toilets and other domestic sullage (collectively known as wastewater).

The solids settle out and break down in the tank. The liquid remains in the tank for a short time before overflowing into a sealed soak away or drain field where it infiltrates into the ground. A permeable soil is essential for the soak away to function properly.

Septic tanks must be emptied periodically (for instance, every three years) and the solids disposed of hygienically. This is usually done with a vacuum tanker.

Septic tanks have the advantages of little maintenance, isolation and partial treatment of excreta, few odour or fly problems, and the possibility of subsequent connection to a sewerage system. Their disadvantages are high cost of construction; the need for periodic mechanical emptying; the need for large volumes of flushing water; and the fact that soak ways can overflow if not designed, built and operated properly. Septic tanks are only suitable where flush toilets are used and there are reasonably large amounts of domestic sullage to be disposed of. They are therefore most likely to be built where there is a house connection to a water supply.

Septic tanks produce liquid effluent which must be disposed of by infiltrating into the ground. The liquid effluent contains large numbers of germs which are dangerous. It is important that they are only disposed of by infiltration, thus septic tanks should only be built where the soil is permeable and the liquid effluent they produce will infiltrate. Where soils are very permeable, however, there is a risk of groundwater contamination from soakaways, particularly where the water table is high.

Septic tanks should only take sewage. Rainwater should go into soakaways. Grease from Kitchen drains should be collected in a grease trap gully because it can clog up the septic tank and stop its effective operation. It is essential to clean gully traps regularly.

Septic tank location.

The location of a septic tank depends on these principles:

- Access for pumping it out needs to be convenient
- It should be at least 15 Meters from a source of water supply.
- It should be at least 3m from the nearest building.
- Avoid areas where rainwater would stand or flow over the tank or vehicles could drive over it.
- It should be downhill from source of sewage
- The effluent should not discharge into water supplies or streams.

d) BIODIGESTER

A **biodigester** is a modern, environmentally friendly alternative to the traditional septic tank. It is an enclosed, oxygen-free (anaerobic) system that uses specialized bacteria to break down organic waste (sewage) into treated water and biogas.

In the Kenyan construction market, biodigesters have become increasingly popular for residential and commercial projects because they solve many of the space and maintenance issues associated with older systems.

How a Biodigester Works

The process relies on **anaerobic digestion**. Unlike a septic tank, which primarily relies on gravity to settle solids, a biodigester actively "digests" the waste.

1. **Inlet:** Raw sewage enters the first chamber of the digester.
2. **Digestion:** Inside, colonies of anaerobic bacteria (often supplemented with a "starter" bacterial culture) feed on the organic matter.
3. **Separation:** This process breaks down solids into liquid effluent and gas.
4. **Outlet:** The treated water exits the system. Because it has been biologically treated, it is much cleaner than septic tank effluent and is often used for gardening or directed into a small soakaway.

Products of a Biodigester

A unique feature of this system is that it converts "waste" into "resources":

- **Treated Water:** This can be used for irrigation, flushing toilets, or safely discharged into the ground.
- **Biogas:** In larger systems, the methane produced during digestion can be captured and used as fuel for cooking or lighting.
- **Bio-slurry:** The small amount of solid remains can be used as a high-quality organic fertilizer.

3.Biodigester vs. Septic Tank

Feature	**Biodigester**	**Septic Tank**
Size	Small and compact (saves space).	Large and bulky.
Maintenance	Rarely needs exhausting (desludging).	Needs exhausting every 2–5 years.
Smell	Odorless if functioning correctly.	Can produce foul odors.
Cost	Higher initial cost, lower maintenance.	Lower initial cost, higher maintenance.
Install Time	Fast (usually 1–2 days).	Long (requires extensive masonry).

4. Why it is popular in Kenya

- **Space Efficiency:** In urban areas like **Syokimau, Kitengela, or Ruiru**, where plot sizes are often 50x100ft, a biodigester takes up very little room, leaving more space for the house or a garden.
- **High Water Table:** In areas where the ground is often wet, traditional septic tanks can fail or "float." Biodigesters are sealed and more resilient.
- **NEMA Compliance:** Because the effluent is cleaner, it is easier to meet the environmental standards set by the **National Environment Management Authority (NEMA)**.

5. Maintenance Tips

To keep the bacteria healthy, you must avoid:

- **Harsh Chemicals:** Strong bleaches or "acid" cleaners can kill the beneficial bacteria.
- **Non-Biodegradables:** Plastics, sanitary pads, and cigarette butts cannot be digested and will cause blockages.
- **Overloading:** Ensure the system is sized correctly for the number of people living in the house.